Thermodynamics of CO_2-loaded aqueous amines with ePC-SAFT. Part I: monoethanolamine calibration for phase equilibria, speciation, absorption heat, density and phase stability

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ABSTRACT

Can a common electrolyte perturbed-chain statistical associating fluid theory (ePC-SAFT) formulation describe loaded aqueous monoethanolamine while keeping literature reaction constants and avoiding a predicted additional liquid? A joint calibration with eleven adjustable parameters and tied amine-ion Born diameters preserves pressure, speciation, caloric and density agreement without a phase-stability penalty. For nominal 30 wt% monoethanolamine, fitted carbon-dioxide pressure average absolute relative deviation is 11.64 %, integral absorption-heat root-mean-square error is 1.98 kJ mol^{-1} carbon dioxide, and fitted loaded-density deviation is 0.152 % without volume translation. The parameter restrictions have a measurable cost: pressure deviation on the same Ma'mun observations rises from 19.43 to 33.47 % relative to the less restricted calibration. Finite searches detect no additional-liquid instability for the adopted parameters; an unpenalized comparison fit instead predicts an ion-rich, water-poor liquid through a coupled energy balance with a prominent dispersion change. Concentration transfer, carbonate shares and high-temperature absorption heat remain inaccurate. All comparisons use data seen during development, and parameter uncertainty is conditional on the fit. The results support a common thermodynamic calculation and a bounded calibration, rather than independent validation or a global stability proof. Part I supplies the equations and shared inputs for the companion study of other amines.

1. Introduction

Can one equation of state describe the equilibrium, caloric and density properties of loaded aqueous monoethanolamine (MEA) with literature reaction constants and fewer adjustable parameters, without predicting an additional liquid? This question matters because absorption pressure, species concentrations, heat release and liquid density enter different parts of an absorber calculation but describe the same reacting solvent [1, 2, 3]. A common Helmholtz energy relates these properties through composition, temperature and volume derivatives. Agreement with one property alone does not establish agreement with the others or the correct number of phases.

1.1. Models of loaded aqueous MEA

Activity-coefficient models combine reaction equilibria with a description of nonideal liquid activities. Böttinger et al. fitted a Pitzer model to species observations while keeping the Bates–Pinching MEA dissociation law [4, Sects. 4–6, Table 5]. Zhang et al. jointly estimated electrolyte non-random two-liquid (eNRTL) parameters from pressure, heat, heat capacity and nuclear magnetic resonance (NMR) data, and compared additional pressure sources separately [1, Sect. 3.3, Tables 9–11]. These studies establish multi-property fitting and explicit accounting of data use as relevant precedents.

Statistical associating fluid theory (SAFT) provides an equation of state for both phases and calculates density directly. Baygi and Pahlavanzadeh combined perturbed-chain SAFT (PC-SAFT) with ideal-activity reaction equilibria after fitting pure-MEA and binary parameters [5, Sects. 3.1–3.3]. Najafloo and Zarei used Huang–Radosz SAFT with the same broad separation between molecular phase properties and ideal reaction activities [6, Sects. 2–3.3]. Reactive-association approaches instead represent reaction products through bonded fractions, including soft-SAFT and group-contribution SAFT-γ Mie [7, 8]. Their species definitions and fitting roles differ from explicit-ion reaction models.

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Electrolyte PC-SAFT (ePC-SAFT) adds ionic electrostatics to the molecular equation of state [9, 10]. Applications to N-methyldiethanolamine (MDEA) estimate subsystem inputs before calculating reacting-mixture pressure and speciation [11, Sects. 4.1–4.3][12, Sects. 6.1–6.3]. Cleeton et al. extend this treatment to competitive acid-gas absorption and discuss its concentration-dependent discrepancies [13, Sects. 3–4]. MDEA does not form carbamate, so this experience does not determine the MEA-ion parameters.

Polley and Hedengren already describe the explicit MEA ions, including carbamate, with ePC-SAFT [14, Sect. 2.2, App. B]. Their parameters were estimated in separate steps, and their column calculations used empirical caloric properties. The present question concerns joint property estimation and phase assessment with one thermodynamic property calculation, rather than the first introduction of explicit MEA ions.

1.2. Property constraints and phase stability

Pressure, species and caloric measurements constrain different combinations of reaction equilibria and physical interactions. NMR may report molecular and protonated MEA as a sum, while titration and carbon-balance estimates give different information about bicarbonate and carbonate [4, Table 3][15, Eqs. 18–19b]. Direct calorimetry also differs from heat inferred from the temperature dependence of pressure [16]. A joint fit must preserve these observable definitions rather than treat every reported value as an interchangeable species fraction or heat.

A single-liquid property objective contains no requirement that the calculated liquid be stable against formation of another phase. Michelsen's tangent-plane criterion tests whether an additional phase can lower Gibbs energy [17, Sect. 2]. For reacting electrolytes the trial must also satisfy charge neutrality and the chemical constraints [18, 19, 20]. Phase searches therefore supply a separate numerical check on a calibration, with a finite search giving no global proof.

1.3. Aim and scope of the series

This work finds that fixed literature reaction laws and Born diameters tied to the amine-ion segment diameters permit a reduced joint calibration with no detected additional-liquid instability in the specified searches. The restrictions preserve useful multi-property agreement but worsen some pressure comparisons and leave concentration transfer, minor species and high-temperature heat discrepancies. The paper defines the ePC-SAFT contributions, reports the cost of the parameter choices, and compares the adopted parameter set with the earlier calibration and published models on identified observations. Data seen during development remain comparisons of calibration and calculations with fixed parameters, not independent validation. Part I establishes the MEA formulation and the exact shared inputs; Part II examines their use with other amines.

2. Thermodynamic model

The liquid contains nine true species: CO_2, MEA, H_2O, $MEAH^+$, $MEACOO^-$, $HCO_3{}^-$, $CO_3{}^{2-}$, H_3O^+ and OH^-. Pressure calculations use a reactive bubble state of one liquid and one incipient vapor at the measured temperature and apparent composition. Species comparisons use single-liquid states at 101 325 Pa. Only the neutral species CO_2, MEA and H_2O enter the vapor, and the calculated CO_2 partial pressure is $p_{CO_2} = y_{CO_2} P$. The same ePC-SAFT equation of state describes both phases.

2.1. ePC-SAFT contributions

The reduced residual Helmholtz energy, $a^{\mathrm{res}} = A^{\mathrm{res}}/(nRT)$, is the sum of hard-chain, dispersion, association, Debye–Hückel and Born contributions [21, 22, 9, 10, 23]:

$$a^{\mathrm{res}} = a^{\mathrm{hc}} + a^{\mathrm{disp}} + a^{\mathrm{assoc}} + a^{\mathrm{DH}} + a^{\mathrm{Born}}. \tag{1}$$

Each neutral species is modeled as a chain of m_i^{seg} spherical segments of diameter σ_i and dispersion energy ϵ_i (tabulated as ϵ_i/k_B), and each ion as a single charged sphere with $m_i^{\mathrm{seg}} = 1$. The following equations define the contributions used throughout this series; relative permittivity supplies the electrostatic closure rather than an additional term in the sum. The pressure and chemical potentials follow from density and composition derivatives of a^{res} [21, Eqs. A.22, A.32 and A.33], and caloric properties from its temperature derivatives (Section 2.3). Because the permittivity in the two electrostatic terms depends on density, temperature and composition (Eq. (18)), these derivatives include its variation. Dispersion energies ϵ_i and ϵ_{ij} are distinct from the association energies ϵ^{AB}; relative permittivities use ε. Below, $\mathcal{I}$ is the set of ions and $\rho_N = N_{\mathrm{A}}\rho$ is number density, with N_{A} the Avogadro constant.

2.1.1. Segment diameters and dispersion interactions

For the neutral species the hard-sphere diameter depends on temperature [21, Eq. A.9],

$$d_i = \sigma_i \left[1 - 0.12 \exp\left(-\frac{3\epsilon_i}{k_B T}\right)\right], \tag{2}$$

with the water diameter law $\sigma_w(T)$ of Section S1 in place of a constant σ_i. The ion packing diameter is the temperature-independent value $d_i = 0.88\sigma_i$; the Debye–Hückel diameter equals σ_i for all six ions, and the Born diameter equals the segment diameter for $MEAH^+$ and $MEACOO^-$. The other four Born diameters remain fixed inputs. These diameter choices define the present parameterization; they are not independently measured ion sizes.

The unlike-pair diameter is $\sigma_{ij} = (\sigma_i + \sigma_j)/2$, and the dispersion energy of a pair is

$$\epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j}\,\left(1 - k_{ij}\right). \tag{3}$$

Dispersion between ions of the same charge sign, including an ion with itself, is excluded ($\epsilon_{ij} = 0$ for $z_i z_j > 0$), while ion–solvent and cation–anion pairs keep Eq. (3) [23]. The CO_2–water, MEA–water and $MEAH^+$–water interactions have the reciprocal-temperature form

$$k_{ij}(T) = k_{ij}(T_{\mathrm{ref}}) + b_{ij}(T^{-1} - T_{\mathrm{ref}}^{-1}), \qquad T_{\mathrm{ref}} = 313.15\,\mathrm{K}, \tag{4}$$

with b_{ij} in kelvin. The CO_2–water coefficients are fixed inputs, whereas the MEA–water and $MEAH^+$–water intercepts and slopes enter the joint fit.

2.1.2. Association scheme

Water and MEA each carry one donor and one acceptor site (2B), and only donor–acceptor pairs bond. CO_2 carries one donor and one acceptor induced-association site, each bonding only with the water site of the opposite role: it does not self-associate, and the CO_2–water bonds use $\epsilon^{AB} = \epsilon_w^{AB}/2$ and $\kappa^{AB} = \kappa_w^{AB}$ (Table 7). This induced-association topology [24] follows Schick et al. and Pabsch et al. [25, Table 1, Sect. 2.4.2][26], with water inputs fitted to IAPWS-95 (Section 4.1). MEA–water cross-association uses the Wolbach–Sandler rule [5, Eqs. 6 and 7][24, Eqs. 7 and 8],

$$\epsilon_{ij}^{AB} = \tfrac{1}{2}\left(\epsilon_i^{AB} + \epsilon_j^{AB}\right), \qquad \kappa_{ij}^{AB} = \sqrt{\kappa_i^{AB}\kappa_j^{AB}}\left(\frac{\sqrt{\sigma_i\sigma_j}}{\sigma_{ij}}\right)^3, \tag{5}$$

and MEA and CO_2 do not bond with each other. The MEA parameters use Baygi and Pahlavanzadeh's 2B model, with its association volume fitted to their model for the σ_{ij}^3 convention of the association strength used here (Eq. 13 and Section S1).

2.1.3. Hard-chain contribution

The hard-chain term, from Wertheim's first-order perturbation theory with the hard-sphere mixture expressions of Boublík and Mansoori et al., is [21, Eqs. A.4–A.8]:

$$a^{\mathrm{hc}} = \bar{m}\, a^{\mathrm{hs}} - \sum_i x_i \left(m_i^{\mathrm{seg}} - 1\right) \ln g_{ii}^{\mathrm{hs}}, \qquad \bar{m} = \sum_i x_i m_i^{\mathrm{seg}}, \tag{6}$$

$$a^{\mathrm{hs}} = \frac{1}{\zeta_0}\left[\frac{3\zeta_1\zeta_2}{1-\zeta_3} + \frac{\zeta_2^3}{\zeta_3\left(1-\zeta_3\right)^2} + \left(\frac{\zeta_2^3}{\zeta_3^2} - \zeta_0\right)\ln\left(1-\zeta_3\right)\right], \tag{7}$$

$$g_{ij}^{\mathrm{hs}} = \frac{1}{1-\zeta_3} + \frac{d_i d_j}{d_i + d_j}\frac{3\zeta_2}{\left(1-\zeta_3\right)^2} + \left(\frac{d_i d_j}{d_i + d_j}\right)^2 \frac{2\zeta_2^2}{\left(1-\zeta_3\right)^3}, \qquad \zeta_n = \frac{\pi}{6}\rho_N \sum_i x_i m_i^{\mathrm{seg}} d_i^n, \quad n = 0, \ldots, 3. \tag{8}$$

Here d_i is the hard-sphere diameter, g_{ij}^{hs} is the hard-sphere pair distribution function at contact, and $\eta = \zeta_3$ is the packing fraction.

2.1.4. Dispersion contribution

The dispersion term is Gross and Sadowski's second-order perturbation expansion [21, Eqs. A.10–A.19, Table 1]:

$$a^{\mathrm{disp}} = -2\pi\rho_N I_1 \sum_i \sum_j x_i x_j m_i^{\mathrm{seg}} m_j^{\mathrm{seg}} \frac{\epsilon_{ij}}{k_B T}\sigma_{ij}^3 - \pi\rho_N \bar{m}\, C_1 I_2 \sum_i \sum_j x_i x_j m_i^{\mathrm{seg}} m_j^{\mathrm{seg}} \left(\frac{\epsilon_{ij}}{k_B T}\right)^2 \sigma_{ij}^3, \tag{9}$$

$$C_1 = \left[1 + \bar{m}\frac{8\eta - 2\eta^2}{(1-\eta)^4} + (1-\bar{m})\,\frac{20\eta - 27\eta^2 + 12\eta^3 - 2\eta^4}{[(1-\eta)(2-\eta)]^2}\right]^{-1}, \tag{10}$$

$$I_1 = \sum_{k=0}^{6} a_k(\bar{m})\,\eta^k, \qquad I_2 = \sum_{k=0}^{6} b_k(\bar{m})\,\eta^k, \qquad a_k(\bar{m}) = a_{0k} + \frac{\bar{m}-1}{\bar{m}}a_{1k} + \frac{\bar{m}-1}{\bar{m}}\frac{\bar{m}-2}{\bar{m}}a_{2k}, \tag{11}$$

where C_1 comes from the hard-chain compressibility, and $b_k(\bar{m})$ has the same form as $a_k(\bar{m})$. The coefficients a_{0k}–a_{2k} and b_{0k}–b_{2k} are Gross and Sadowski's universal constants. The unlike-pair diameter is $\sigma_{ij} = (\sigma_i + \sigma_j)/2$, and ϵ_{ij} is the pair dispersion energy of Eq. 3.

2.1.5. Association contribution

Hydrogen bonding is described by Wertheim's first-order theory in the form used with PC-SAFT [24, Eqs. 4–6]:

$$a^{\mathrm{assoc}} = \sum_i x_i \sum_{A\in i}\left(\ln X_A - \frac{X_A}{2} + \frac{1}{2}\right), \qquad X_A = \left[1 + \rho_N \sum_j x_j \sum_{B\in j} X_B \Delta^{AB}\right]^{-1}, \tag{12}$$

$$\Delta^{AB} = g_{ij}^{\mathrm{hs}}\,\sigma_{ij}^3\,\kappa^{AB}\left[\exp\left(\frac{\epsilon^{AB}}{k_B T}\right) - 1\right], \tag{13}$$

where X_A is the fraction of molecules i not bonded at site A, B runs over the sites on molecule j, and Δ^{AB} is the strength of an A–B bond between molecules i and j. The site fractions are solved at every state. The site scheme and the MEA–water combining rule are given in the association scheme above.

2.1.6. Debye–Hückel contribution

The Debye–Hückel term of Cameretti et al. describes ions of diameter d_i^{DH} in a continuum of relative permittivity $\varepsilon_{\mathrm{bulk}}$ [9, Eqs. 5, 13 and 14]:

$$a^{\mathrm{DH}} = -\frac{\kappa_{\mathrm{D}} e^2}{12\pi\varepsilon_0\varepsilon_{\mathrm{bulk}} k_B T}\sum_{i\in\mathcal{I}} x_i z_i^2 \chi_i, \qquad \kappa_{\mathrm{D}}^2 = \frac{e^2\rho_N}{\varepsilon_0\varepsilon_{\mathrm{bulk}} k_B T}\sum_{i\in\mathcal{I}} x_i z_i^2, \tag{14}$$

$$\chi_i = \frac{3}{\left(\kappa_{\mathrm{D}} d_i^{\mathrm{DH}}\right)^3}\left[\frac{3}{2} + \ln\left(1 + \kappa_{\mathrm{D}} d_i^{\mathrm{DH}}\right) - 2\left(1 + \kappa_{\mathrm{D}} d_i^{\mathrm{DH}}\right) + \frac{1}{2}\left(1 + \kappa_{\mathrm{D}} d_i^{\mathrm{DH}}\right)^2\right]. \tag{15}$$

Here κ_{D} is the inverse Debye screening length, with ρ_N in SI units, and χ_i accounts for the finite ion size; $\chi_i \to 1$ as $\kappa_{\mathrm{D}} d_i^{\mathrm{DH}} \to 0$, which recovers the Debye–Hückel limiting law.

2.1.7. Born contribution with solvation shell and dielectric saturation

The Born term follows the solvation-shell and dielectric-saturation (SSM+DS) form of Figiel et al. [23, Eqs. 5–10], with its Eq. 7 as corrected in 2026 [27]:

$$a^{\mathrm{Born}} = -\frac{e^2}{4\pi\varepsilon_0 k_{\mathrm{B}} T}\sum_{i\in\mathcal{I}} x_i z_i^2\left[\frac{1 - \varepsilon_{\mathrm{bulk}}^{-1}}{D_i} + c_{\mathrm{dielectric}}\left(1 - \varepsilon_{\mathrm{ion}}^{-1}\right)\left(\frac{1}{d_i^{\mathrm{Born}}} - \frac{1}{D_i}\right)\right], \tag{16}$$

$$D_i = d_i^{\mathrm{Born}} \left[1 + c_{\mathrm{shell}} \frac{f_{\mathrm{mix}} - 1}{|z_i|} \right], \qquad f_{\mathrm{mix}} = \frac{\sum_k x_k f_k}{\sum_k x_k}. \tag{17}$$

Here d_i^{Born} is the Born diameter, D_i is the outer diameter of the first solvation shell and $\varepsilon_{\mathrm{ion}} = 8$ is the fixed permittivity assigned to the region between the ion surface and that shell. This work uses $c_{\mathrm{shell}} = c_{\mathrm{dielectric}} = 1$, which gives the corrected Eq. 7. The water solvation factor is $f_{\mathrm{H_2O}} = 1.5$ [23, Table 2], and every other component, including MEA, CO_2 and all ions, has $f_k = 1$. Figiel et al. restrict the average in Eq. (17) to ions and solvents; here it runs over all nine components, so $f_{\mathrm{mix}} = 1 + 0.5\, x_{\mathrm{H_2O}}$. The shell diameter therefore changes with loading through the water mole fraction, and the Born term contributes to the water activity.

2.1.8. Relative permittivity

The bulk permittivity in the Debye–Hückel and Born terms depends on solvent density, temperature and neutral composition through the Schreckenberg form reported by Yang et al. [28, Sect. 2.1, Eqs. 1a, 2a and 2b]:

$$\varepsilon_{\mathrm{bulk}} = 1 + \frac{n_s}{V} d_{V,\mathrm{mix}} \left(\frac{d_{T,\mathrm{mix}}}{T} - 1 \right), \qquad d_{q,\mathrm{mix}} = \frac{\sum_{i \in S} n_i d_{q,i}}{n_s}, \quad q \in \{V, T\}. \tag{18}$$

Here $n_s = \sum_{i \in S} n_i$, V is the phase volume, and the neutral set S comprises water, MEA and molecular CO_2. The dielectric-volume and dielectric-temperature coefficients have units of volume per mole and kelvin, respectively, using consistent units for n_s/V. The ion amounts do not enter the neutral mixing averages; their presence affects the phase density and neutral amounts. This law approaches unity as the phase density approaches zero at fixed composition. Its MEA coefficients were fitted to unloaded MEA–water permittivities using equation-of-state densities (Section 4.1); the water and assumed CO_2 coefficients remain fixed. Use of the dielectric law at other temperatures and in loaded solutions is a model assumption.

2.2. Reactions and phase equilibrium

Property calculations use an apparent feed of 1 mol of total amine, introduced as MEA, the water amount given by the source composition, and α mol of CO_2, where α is the loading in mol CO_2 per mol total amine. The water amount follows from the reported MEA molality where the source gives one and otherwise from the reported CO_2-free MEA mass fraction. The liquid conserves the carbon, hydrogen, nitrogen and oxygen amounts of this feed. For these nine species the four element balances imply electroneutrality and leave the five independent reactions of Table 1. Bubble states satisfy equal chemical potentials of CO_2, MEA and H_2O between phases. Every reacting liquid satisfies, for each reaction r,

$$\sum_i \nu_{ri} \ln a_i = \ln K_r(T), \tag{19}$$

where a_i is dimensionless. Writing the equation-of-state fugacity coefficient as $\ln \phi_i = \mu_i^{\mathrm{res}}/(RT) - \ln Z$, with compressibility factor $Z = P/(\rho RT)$, gives

$$a_w = x_w \frac{\phi_w(T, P, \boldsymbol{x}) P}{\phi_w^{\mathrm{pure}}(T, P^\circ) P^\circ}, \qquad a_i = \frac{x_i}{M_w m^\circ} \frac{\phi_i(T, P, \boldsymbol{x}) P}{\phi_i^{\infty, w}(T, P^\circ) P^\circ} \quad (i \neq w). \tag{20}$$

Here M_w is water's molar mass in kg/mol, ρ is molar density and $m^\circ = 1\,\mathrm{mol\,kg^{-1}}$. The infinite-dilution reference is evaluated along a charge-neutral dilution path in pure water; only charge-neutral combinations of species, as they appear in the reactions, require this reference. Thus $a_i = (m_i/m^\circ)\gamma_i^{(m)}$, with $m_i = x_i/(M_w x_w)$ and $\gamma_i^{(m)} = x_w \phi_i P/(\phi_i^{\infty, w} P^\circ)$. A molality-basis constant relates to the corresponding mole-fraction constant by

$$\ln K_{m,r} = \ln K_{x,r} - \Big(\sum_{i \neq \mathrm{H_2O}} \nu_{ri} \Big) \ln \Big(M_{\mathrm{H_2O}} m^\circ \Big), \tag{21}$$

with $m^\circ = 1\,\mathrm{mol\,kg^{-1}}$, so $-\ln(M_{\mathrm{H_2O}} m^\circ) = 4.0165$. The R1–R3 source standard states are at system pressure, and the R4 and R5 constants carry their source standard pressure of 100 kPa.

R1–R3 use the Austgen correlation form [29, Table V],

$$\ln K_r = A_r + B_r/T + C_r \ln T + D_r T + \Delta_r, \tag{22}$$

where Δ_r converts the source to the common aqueous-molality basis. R4 follows Tong et al. [30, Table 5], via Aroua et al. [31]; R5 is Bates and Pinching [32, Eq. 10, Table 2, p. 351], also tabulated by Böttinger et al. [4, Table 5]. All five reaction laws remain fixed in the joint fit. R2 uses Austgen's correlation with its molality conversion, and R5 uses Bates–Pinching's published dissociation correlation; no fitted reaction-enthalpy adjustment is added. The R1–R3 correlation range is 273.15–498.15 K, as stated by Austgen et al. [29, Table V, p. 547, citing Edwards et al. (1978)]. This is the source-correlation range, not verified measurement coverage of the underlying experiments. The reacting calculations in this paper use 293.15–393.15 K; R4 and R5 are extrapolated above their source upper limit of 323.15 K.

Table 1
Completed aqueous-molality laws, $\ln K = A + B/T + C \ln T + DT$, T in K. Source-to-molality offsets are included once. R1–R3: Austgen Table V; R4: Tong Table 5; R5: Bates–Pinching Eq. (10). All laws are fixed in the joint fit.

Reaction	Stoichiometry	A	B (K)	C	D (1/K)
R1	$2\,H_2O \rightleftharpoons H_3O^+ + OH^-$	140.93207	−13445.9	−22.4773	0
R2	$CO_2 + 2\,H_2O \rightleftharpoons HCO_3^- + H_3O^+$	235.4815	−12092.1	−36.7816	0
R3	$HCO_3^- + H_2O \rightleftharpoons CO_3^{2-} + H_3O^+$	220.065535	−12431.7	−35.4819	0
R4	$MEACOO^- + H_2O \rightleftharpoons HCO_3^- + MEA$	2.151	−1545.3	0	0
R5	$MEAH^+ + H_2O \rightleftharpoons MEA + H_3O^+$	−0.890870172	−6166.11565	0	−0.000984815644

2.3. Enthalpy and heat capacity

Total enthalpy follows the same Helmholtz energy and reaction standard states as the equilibrium calculation:

$$H = PV - RT^2 \left(\frac{\partial (A/RT)}{\partial T} \right)_{V,\boldsymbol{n}}, \qquad A = A^{\mathrm{ig}} + A^{\mathrm{res}}. \tag{23}$$

Neutral ideal-gas functions supply the caloric reference; the ionic functions are completed from the five reaction laws, including $\Delta_r h^\circ = RT^2 d \ln K_r / dT$ and the derivatives of the infinite-dilution reference corrections. Reaction enthalpies are therefore included once through the standard-state functions. The neutral ideal-gas coefficients follow NIST-JANAF and Zhang et al. [33, 1] and are listed in the data deposit; Section S4 gives the functions and reference convention, including the water ideal-gas fit to IAPWS-95 over 273.15–400 K. Equilibrium isobaric heat capacity is $C_P = (dH_{\mathrm{eq}}/dT)_{P,\boldsymbol{n}_{\mathrm{feed}}}$, including the change in speciation. The objective uses C_P/n_{feed} per mole of complete analytical feed, with source mass-specific values converted on that same basis. The mass-specific value is $c_P = C_P/m_{\mathrm{feed}}$, where $m_{\mathrm{feed}} = n_{\mathrm{MEA}} M_{\mathrm{MEA}} + n_w M_w + n_{\mathrm{CO_2}} M_{\mathrm{CO_2}}$ is the complete analytical-feed mass.

3. Data and parameter estimation

3.1. Experimental data and their use in fitting

Loading is $\alpha = n_{\mathrm{CO_2,added}}/n_{\mathrm{amine,total}}$, and unloaded-solvent mass fraction is $w_{\mathrm{MEA}} = m_{\mathrm{MEA}}/(m_{\mathrm{MEA}} + m_{\mathrm{H_2O}})$. Three data-role terms are used throughout: *fitted* means an observation enters the objective of the reported fit; *not fitted* means it is outside that objective; *comparison only* means a different quantity or different conditions from those the model was built for, never used to choose a fit. Prior inspection or use in choosing parameters is disclosed separately in Table 2. The objective contains 307 quantities; the binary CO_2–water pressures remain in it as checks on a previously fitted, fixed interaction. All subsequent property calculations use the adopted parameters without adjustment.

Table 2
Data roles in the reported joint fit and earlier use. Comparison sets overlap and their counts are not additive. The fixed Kiepe interaction was fitted separately. No present Part I group qualifies as comparison only because all were seen during development.

Quantity	n	Role	Earlier use
Pressure, species and heat; CO_2 partial pressure (Pa)	47	fitted	Joint objective
Pressure, species and heat; species fraction	94	fitted	Joint objective
Pressure, species and heat; integral heat (J/mol CO_2)	8	fitted	Joint objective
Additional species; species fraction	7	fitted	Joint objective
Jakobsen ratios; species / total amine	6	fitted	Joint objective
CO_2–water; pressure (Pa)	39	fitted	Joint objective
CO_2-free density; density (kg m^{-3})	5	fitted	Joint objective
MEA–water; temperature (K)	25	fitted	Joint objective
MEA–water; vapor water fraction	25	fitted	Joint objective
Loaded density; density (kg m^{-3})	47	fitted	Joint objective
Heat capacity; J/(mol K); total analytical feed	4	fitted	Joint objective
Six-source pressures	159	47 fitted; 112 not fitted	Outside-objective rows inspected before adoption
Aronu, other concentrations	70	not fitted	Inspected before adoption
Wagner pressures	23	not fitted	Inspected before adoption
Matin bicarbonate totals	19	not fitted	Inspected before adoption
Jakobsen carbonate shares	10	not fitted	Inspected before adoption
Arcis heats	14	not fitted	Used in parameter selection
Kim finite-interval heats	73	not fitted	Earlier fitting and prior inspection
All density observations	210	52 fitted; 158 not fitted	Source comparisons informed fitting
Broad heat capacities	51	2 fitted; 49 not fitted	Inspected before adoption

Each species observable follows its experimental definition. Böttinger et al. report MEA + $MEAH^+$ as an NMR sum and tabulate bicarbonate while noting practically absent carbonate [4, p. 134, Table 3]. Their bicarbonate-labeled observations are compared with calculated $HCO_3^- + CO_3^{2-}$. Matin et al. neglect carbonate when inferring bicarbonate from titration and carbon balance [15, pp. 6615–6616, Eqs. 18–19b]; the same aggregate comparison is an observation approximation, and these bicarbonate values are not fitted. The Matin series labeled 294.15 K is calculated at 293.15 K. Jakobsen species-to-total-amine ratios enter the objective, whereas their carbonate shares of dissolved carbon do not; the calculation accounts for dilution during NMR preparation [34, Tables A1–A2, Sample preparation].

Pressure and species point calculations use each source's solvent composition, including reported molality and temperature for Hilliard and Xu [35, App. D.4][36, Table 1]. Three Xu pressure observations above 393.15 K and a Böttinger state with quantifiable oxazolidone are excluded; oxazolidone is absent from the model. The pressure sources are Aronu, Hilliard, Idris, Jou, Ma'mun and Xu, with Wagner assessed separately [37, 35, 38, 39, 40, 36, 41]. Carrier gas is omitted from the bubble calculation: calculated $y_{CO_2}P$ is compared with measured CO_2 partial pressure rather than imposing the experimental total pressure. This approximation can matter at low CO_2 pressure; Neumann et al. instead include nitrogen [42, Sect. 4.2, Fig. 5].

3.2. Absorption heat, heat capacity and density

Vinjarapu integral heats are fitted at 313.15 K and 1.734 MPa from initially carbon-free 30 wt% solvent [43, Sects. 1.1.1 and 2.2, Table 4]. Signed heat per mole of added CO_2 is

$$\Delta h = \frac{H_{\mathrm{final}} - H_{\mathrm{initial}} - \Delta n_{CO_2} h_{CO_2}^{\mathrm{feed}}}{\Delta n_{CO_2}}, \tag{24}$$

where both equilibrium liquids and the pure-CO_2 feed are evaluated at the same temperature and pressure; negative values denote exothermic absorption. Arcis integral heats use 322.5 K and the reported pressures [44, Sect. 2.4, Table 5]. Kim–Svendsen semi-differential calorimetry measures finite additions; the corresponding calculation uses

the liquid enthalpy at each interval endpoint, rather than the cumulative heat from zero loading [16, Tables 1–3]. A 1.734 MPa isobar approximates these intervals because their experimental pressure histories are unavailable.

The fitted Hilliard heat capacities use 7 mol MEA/kg water at 101 325 Pa, with loading 10^{-6} for nominally unloaded states [35, Sects. 4.5–4.6, App. G.2]. The objective uses molar-feed heat capacity, while comparisons with the broader heat-capacity data use the source mass basis. The feed basis includes added CO_2 in both conversions.

Densities come directly from the equation of state, without volume translation. The fitted loaded observations comprise Amundsen and Hartono data, while Han densities are not fitted [3, 45, 46]. Five Amundsen observations at loading 0.4 are excluded from fitting after comparison with interpolated Hartono values; they remain in the displayed density comparisons. The loaded-density residual scale is

$$s_{\rho,j} = \left[u_{\rho,j}^2 + (g_j u_{\alpha,j})^2\right]^{1/2}, \tag{25}$$

where g_j is the secant between neighboring loadings at the same source temperature and composition, with a one-sided secant at either end. These heterogeneous scales are not uniformly standard measurement uncertainties, so twice a scale does not uniformly define a confidence interval. Section S3 gives the source inputs and their qualifications.

3.3. Parameter estimation

Estimation proceeds from pure water against IAPWS-95, through the CO_2–water interaction against Kiepe pressures and the MEA dielectric coefficients against unloaded-solution permittivities, to the joint MEA fit. Earlier subsystem coefficients remain fixed in the joint fit. The adopted set, labeled F2 for the parameter-choice comparison, fixes R2 and R5 to their literature laws and ties the $MEAH^+$ and $MEACOO^-$ Born diameters to their segment diameters. Eleven remaining parameters minimize

$$C(\boldsymbol{\theta}) = \frac{1}{2}\sum_j w_j r_j^2, \qquad r_j = \frac{y_j^{\mathrm{calc}} - y_j^{\mathrm{obs}}}{s_j} \quad \text{or} \quad r_j = \frac{\ln(y_j^{\mathrm{calc}}/y_j^{\mathrm{obs}})}{s_j}, \tag{26}$$

with the residual form, scale and weight specified by quantity in Table S3. Bounded Levenberg–Marquardt least squares uses exact equation-of-state and implicit equilibrium derivatives. Two starts are examined for each restriction; the lower-cost converged start is reported, with convergence determined by the optimizer tolerance rather than an iteration limit. These are local solutions, not a global parameter search.

Pressure uses a log-ratio residual with an assigned relative allowance; species fractions and ratios use relative-plus-floor scales. Vinjarapu heat scales come from the reported heat uncertainty, density scales combine density and loading information, and the heat-capacity scale uses Hilliard's stated accuracy on the molar-feed basis [43, Table 4][35, Sect. 4.6]. The binary-temperature, vapor-fraction and carbon-free-density allowances, together with the group weights, are assigned fitting choices. The reasons for their particular magnitudes were not recorded; they are not reconstructed as statistical sampling variances. Thus the objective specifies relative fitting priorities, and its value is not an absolute measure of physical accuracy.

Agreement is reported as average absolute relative deviation (AARD) and root-mean-square error (RMSE),

$$\mathrm{AARD} = \frac{100\,\%}{n}\sum_j \mathrm{abs}\left(y_j^{\mathrm{calc}}/y_j^{\mathrm{obs}} - 1\right), \qquad \mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_j (y_j^{\mathrm{calc}} - y_j^{\mathrm{obs}})^2}. \tag{27}$$

AARD describes error relative to observations; it is not an uncertainty interval. For the interior parameters, local covariance is

$$\mathrm{cov}(\boldsymbol{\theta}_I) = s_C^2 (\boldsymbol{J}_I^{\mathsf{T}} \boldsymbol{J}_I)^{-1}, \qquad s_C^2 = \frac{2C}{N - |I|}, \tag{28}$$

where $\boldsymbol{J}_I$ is the weighted residual Jacobian and active-bound parameters remain fixed. These errors condition on the fitted model, source laws, scales, weights and local linearization.

3.4. Phase-stability analysis and stability penalty

The reacting tangent-plane criterion tests charge-neutral trial phases [17, Sect. 2]:

$$D(\boldsymbol{w}) = g(T, P, \boldsymbol{w})/(RT) - \boldsymbol{w}^{\mathsf{T}} \boldsymbol{B}^{\mathsf{T}} \boldsymbol{\lambda}, \quad \sum_i w_i = 1, \quad \sum_i z_i w_i = 0, \quad w_i > 0. \tag{29}$$

Here g is molar Gibbs energy, $\boldsymbol{B}$ describes conserved component combinations, and $\boldsymbol{\lambda}$ contains their chemical-potential multipliers in the parent equilibrium. A negative distance must be accompanied by a material-conserving finite transfer that lowers Gibbs energy beyond its numerical error estimate before instability is counted. An unavailable or incomplete search supplies no evidence of stability.

Each fitted set is searched on the liquid-state grid and on liquid–vapor states defined in Section S6. The liquid grids use 1.05 and 3.0 times the candidate's bubble pressure; separate checks use 1.5 times that pressure and intermediate compositions and temperatures. At each liquid–vapor state, trial compositions that previously produced a resolved Gibbs decrease are evaluated in addition to optimized trials. These supplied-composition distances are separate checks, not newly resolved phase splits.

If a split is detected, the whole searched liquid-state set enters the penalized objective:

$$r_k^{\mathrm{stab}} = \sqrt{w_{\mathrm{stab}}}\,\frac{\tau_k}{s_{\mathrm{stab}}}, \qquad C_{\mathrm{pen}} = C + \frac{1}{2}\sum_k (r_k^{\mathrm{stab}})^2, \quad w_{\mathrm{stab}} = 1, \quad s_{\mathrm{stab}} \in \{10^{-2}, 3\times 10^{-3}\}. \tag{30}$$

The two scales are successive numerical penalty strengths; τ_k is the most negative distance on a resolved descending branch, or zero when a completed search finds none. No zero is assigned to an unavailable search. If needed, a parameter segment from a separately admissible anchor to the penalized solution is shortened in 0.01 steps, then refined by ten bisections to recover an admissible endpoint. The adopted F2 fit required neither a penalty stage nor this retreat.

Table 3 supplies reproduction values for a CO_2–water, an MEA–water and a loaded-MEA state. The two initial-pressure solutions agree at the displayed precision; Section S4 lists the corresponding energy contributions and species fractions. These are numerical reproduction checks, not additional measurements or physical validation.

Table 3
Bubble-state reproduction values. Inputs: $x_{CO_2} = 0.00036$ in water; $x_{MEA} = 0.9264$ in MEA–water; 30 wt% MEA at loading 0.3 mol/mol. Each was solved from two initial pressures; printed values agree. Density is liquid mass density.

System	T (K)	P (Pa)	ρ (kg/m^3)
CO_2–water	313.2	96620.50005	992.6594988
MEA–water	350	3983.444831	972.7825398
Loaded MEA	313.15	6639.529794	1060.792115

4. Results and discussion

4.1. Pure water and binary subsystems

The pure-water fit reproduces IAPWS-95 saturation density, vapor pressure and heat capacity over the fitted interval, with interleaved temperatures providing a numerical check of the correlation [47]. The water inputs remain fixed during MEA estimation; the comparison is with a reference equation rather than independent measurements. Section S1 gives the water diameter law and the source of each fixed species input.

The CO_2–water interaction remains fixed after estimation against Kiepe pressures, while the MEA–water interaction is jointly estimated with the amine-ion parameters [48, 49]. Their errors are reported by observable in Table 5. The MEA–water temperature–composition curves show the binary constraint on that interaction before the reacting ternary mixture is considered (Fig. 1). The dielectric coefficients were estimated from unloaded solutions at 298.15 K; these observations do not determine the temperature or loading dependence used in the reacting mixture [50, ESI Tables S1–S2].

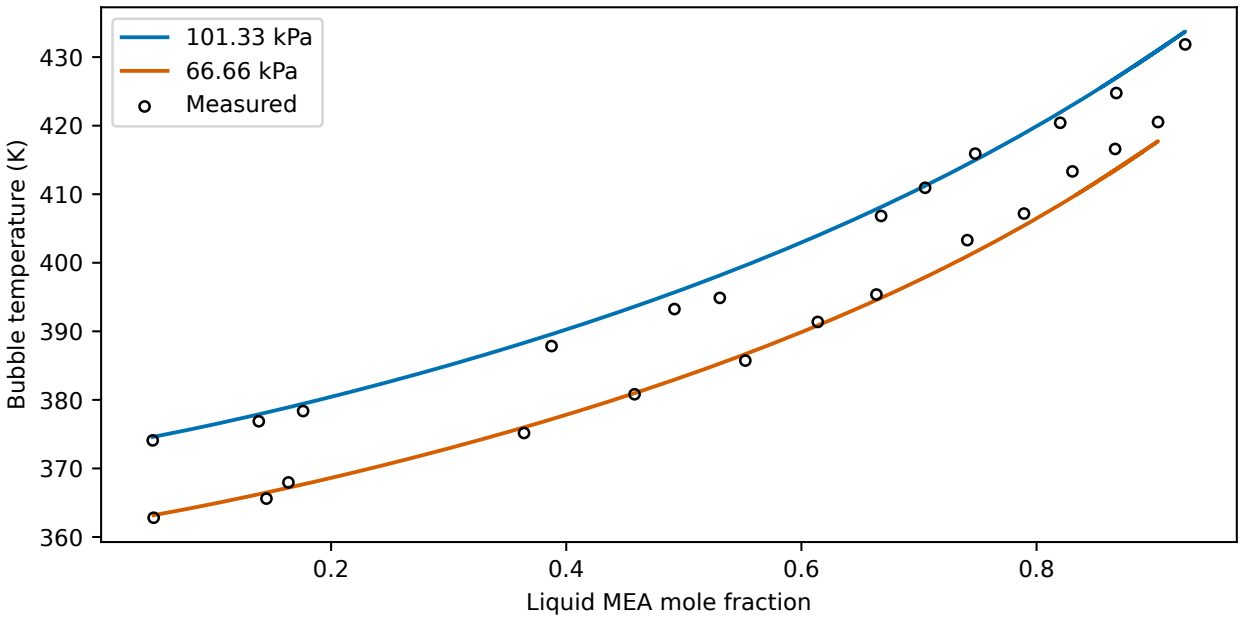


Figure 1: MEA–water bubble temperature against liquid MEA mole fraction. Lines use the adopted parameters at the labeled pressures; circles are the fitted Cai observations [49].

4.2. Phase stability of the fitted models

Loaded aqueous MEA is observed as a homogeneous liquid in Böttinger et al.'s measured subset [4, Sect. 3]. That observation motivates checking the model for a spurious additional liquid, although it does not establish experimental homogeneity over the entire numerical search domain. The adopted F2 parameters give no detected instability at 1908 liquid-grid states, 576 intermediate liquid states and 162 liquid–vapor states, with no unavailable or incomplete search in these sets. Trial compositions from previously resolved descents also give no negative distance; the final check covers 6121 such compositions at every liquid–vapor state. These results required no stability penalty during F2 fitting.

The unpenalized 16-parameter comparison fit gives a resolved Gibbs decrease at 54 of the same 162 liquid–vapor states. It is distinct from F0, the stability-penalized calibration in the parameter-choice comparison. Its strongest confirmed split, at 333.15 K, loading 0.5 mol/mol and 21.175 719 kPa, produces a water-poor, ion-rich liquid alongside a water-rich liquid and vapor. The calculated water mole fractions are 0.590900 and 0.973235 in the two liquids; these are model compositions, not measured second-liquid compositions. Figure 2 compares the energy balance for this split under the unpenalized parameters and F2. Phase amounts and compositions remain frozen for this comparison, while densities are recalculated at the imposed pressure.

The Born contribution is strongly favorable to the illustrated split and becomes more favorable with F2, while dispersion supplies the largest upward change. Across the confirmed configurations, dispersion has the largest absolute change in 49 of 51 cases; Born has the largest change in the other two. The split therefore reflects a coupled energy balance with a prominent dispersion change, rather than one uniquely identified microscopic cause. A negative energy change relative to the old parent is not itself an F2 instability test because that parent is not an equilibrated F2 state.

Two distinct isolated changes remove all detected descents in the finite liquid–vapor search: replacing only the $MEAH^+$ segment diameter by its F2 value, or replacing only the $MEAH^+$–water interaction constant. In the segment-diameter probe, the Debye–Hückel, packing and Born diameters remain at the unpenalized fit's values. These probes lie outside the tied parameterization of F2, and neither is a fitted parameter set with assessed property agreement. They show that the sufficient parameter change is not unique; they do not identify a molecular mechanism or uncertainty in a stability boundary.

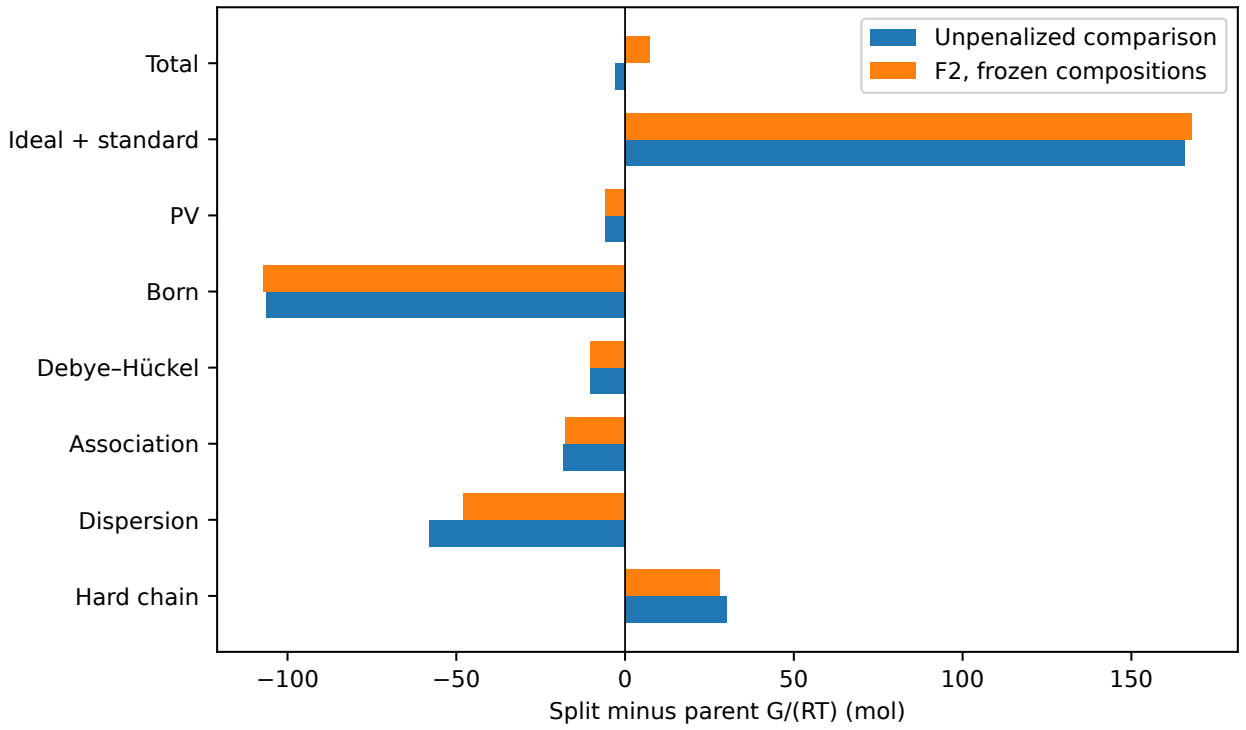


Figure 2: The false liquid split involves opposing energy contributions. Extensive split-minus-parent Gibbs contributions at 333.15 K, loading 0.5 mol/mol and 21.175 719 kPa, for the unpenalized comparison fit and F2 evaluated at the same frozen phase amounts and compositions. Densities satisfy the imposed pressure separately for each parameter set. $G/(RT)$ has units of mol on the feed basis of 9.7292821 mol MEA, 76.9751925 mol water and 4.86464105 mol CO_2. The F2 bars are diagnostic evaluations, not equilibrated F2 split phases.

4.3. CO_2–MEA–water at 30 wt%

4.3.1. Pressure and speciation

Figure 3 shows the pressure–loading relation with source observations, and Table 4 gives the source-specific errors. The fitted pressures constrain the lower-temperature part of this comparison; the higher-temperature pressures are calculated without further fitting. The deviations therefore describe distinct data roles even where the observations appear on the same curve. Source differences and the strong loading dependence remain visible despite agreement over the broad logarithmic pressure range.

Table 4
Pressure errors by experimental source at nominal 30 wt% MEA; source subsets contain both fitted observations and observations that are not fitted.

Source	n	AARD (percent)
Aronu et al. (2011)	36	31.2
Hilliard (2008)	31	9.245
Idris et al. (2014)	10	20.97
Jou et al. (1995)	48	14.41
Ma'mun et al. (2005)	19	33.47
Xu and Rochelle (2011)	15	25.81

The species curves show molecular MEA being replaced by protonated MEA and carbamate as loading increases, followed by increasing bicarbonate (Fig. 4). Agreement with the NMR amine sum does not independently establish the separate molecular and protonated fractions. The carbonate share remains far below Jakobsen's observations, even

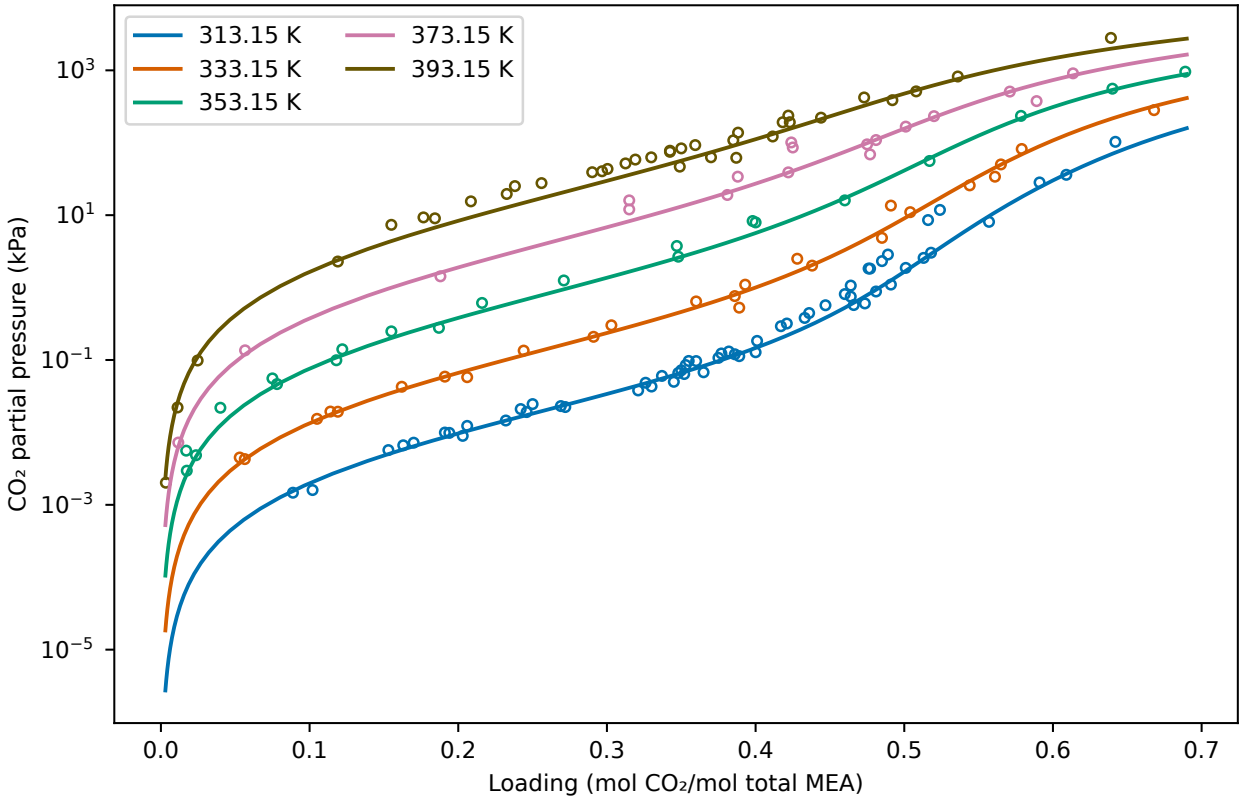


Figure 3: Measured CO_2 partial pressures and F2 reactive-bubble curves at nominal 30 wt% MEA. Circles are six-source observations assigned to the labeled nominal temperature within 2 K; lines use nominal compositions, while source-specific point calculations preserve reported compositions and temperatures. Counts for the fitted and not fitted observations, with their prior uses, are in Table 2; source errors are in Table 4.

Table 5
Fitted observations by quantity. AARD is in percent; RMSE uses the stated unit. These groups are disjoint.

Quantity (unit)	n	AARD	RMSE
Pressure, species and heat; CO_2 partial pressure (Pa)	47	11.64	7495
Pressure, species and heat; species fraction	94	7.079	0.002683
Pressure, species and heat; integral heat (J/mol CO_2)	8	1.969	1979
Additional species; species fraction	7	8.928	0.003463
Jakobsen ratios; species / total amine	6	17.07	0.04612
CO_2–water; pressure (Pa)	39	2.667	1.8×10^5
CO_2-free density; density (kg m^{-3})	5	0.03036	0.3622
MEA–water; temperature (K)	25	0.3695	1.785
MEA–water; vapor water fraction	25	2.602	0.02036
Loaded density; density (kg m^{-3})	47	0.1523	1.842
Heat capacity; J/(mol K); total analytical feed	4	2.835	3.003

though the total-amine ratios are fitted; the two observables constrain different combinations of species. The fitted pressures do not determine that minor-species partition uniquely.

Carbonate formation depends on its reaction constant and relative ionic activities, so a small carbonate share cannot be attributed to a single parameter from these comparisons. Titration-derived bicarbonate totals also differ from the NMR observations, as Matin et al. discuss [15, pp. 6616–6617]. The experimental observable approximations and

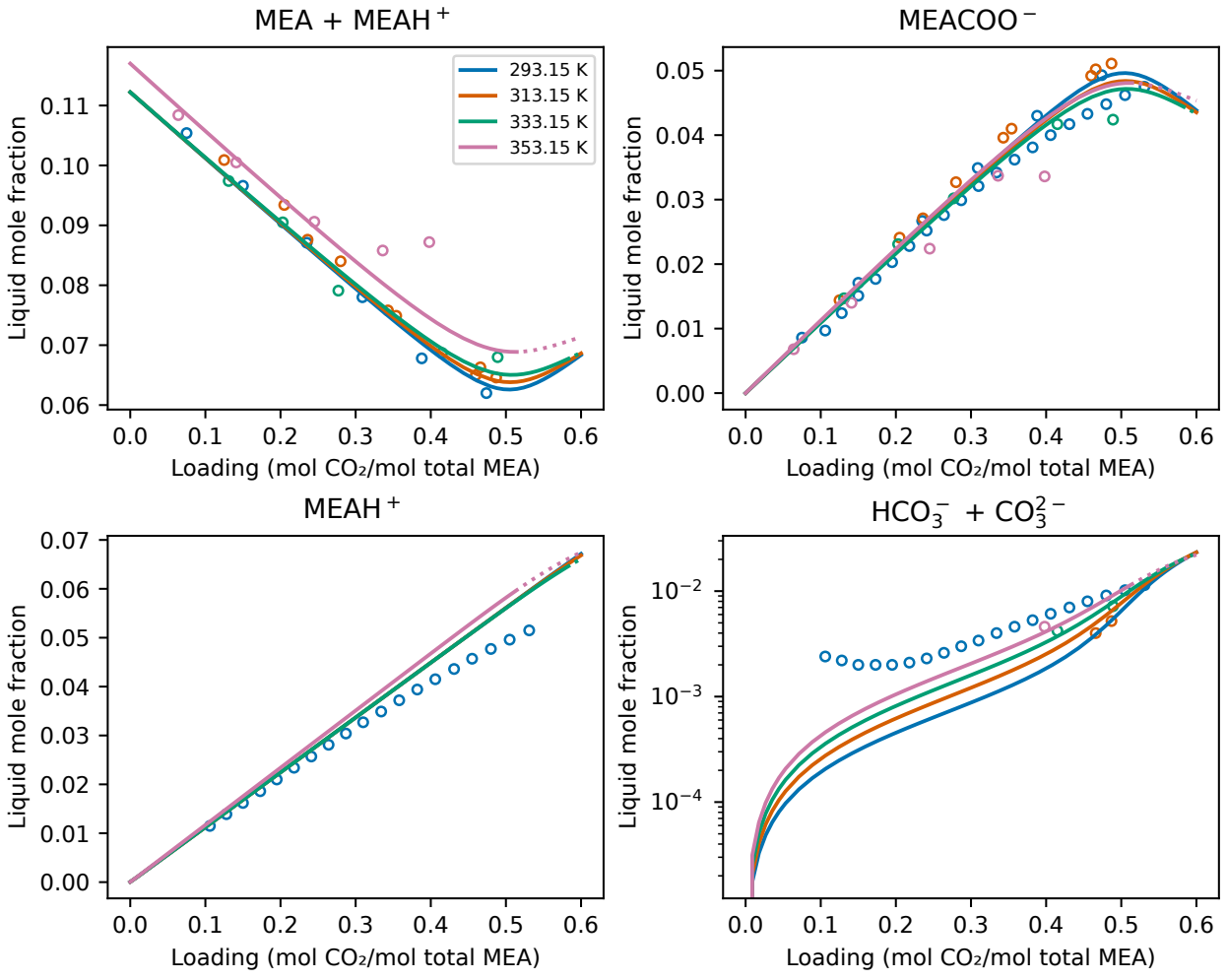


Figure 4: Species and aggregate liquid mole fractions against loading at the labeled temperatures. Solid lines are F2 single-liquid calculations at 101 325 Pa; dotted portions lie below their calculated bubble pressure. Circles combine Böttinger NMR and Matin titration observations with the aggregate definitions in Section 3 [4, 15]. The highest-temperature NMR series uses 31 wt% MEA; the others use nominal 30 wt%. All observations are fitted except that NMR series and the Matin bicarbonate totals.

model discrepancy both matter; the available evidence does not isolate their contributions. Table 9 reports carbonate and species errors on the same observations as the predecessor model.

4.3.2. Absorption heat and heat capacity

The fitted integral heats and the Arcis comparison probe different temperature and pressure conditions (Fig. 5). The calculated Arcis absorption is less exothermic than measured, with a larger discrepancy in the lower-pressure series. The measurements themselves differ between those series at nearby loadings [44, Table 5]; neither a pressure-only explanation nor a unique model contribution is isolated by this comparison.

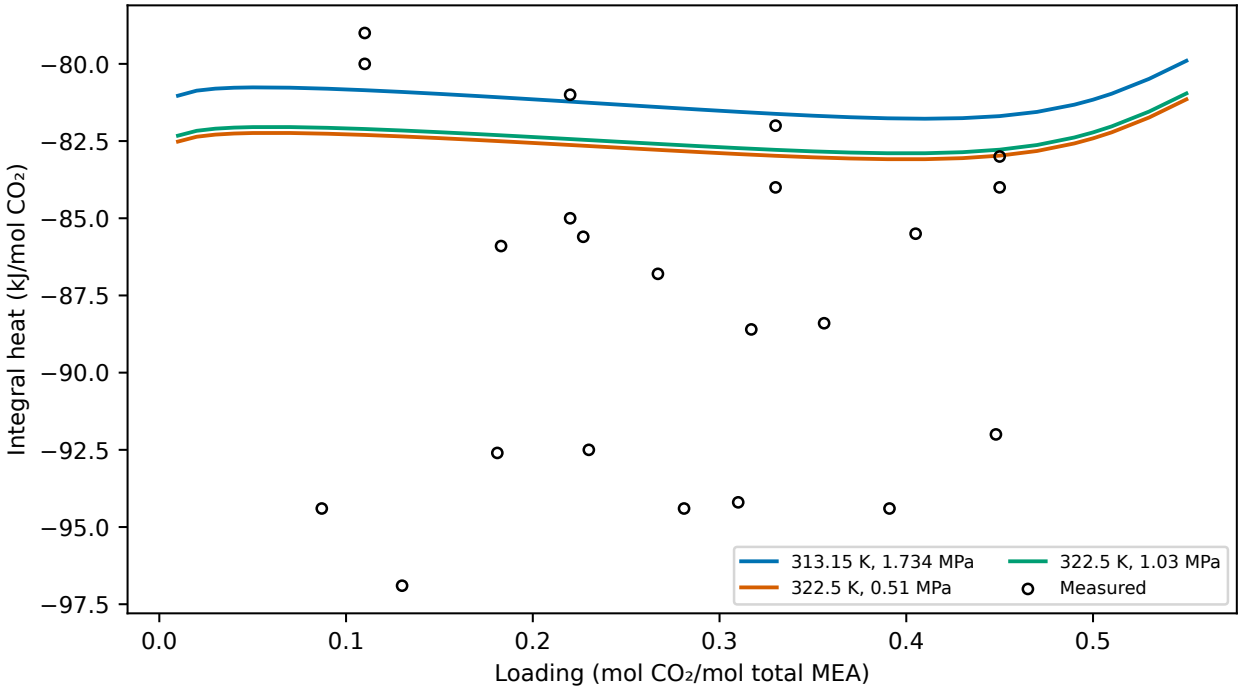


Figure 5: Integral absorption heat from initially carbon-free nominal 30 wt% MEA, per mole of added CO_2. Lines are F2 calculations at the labeled temperature and pressure; circles are fitted Vinjarapu and Arcis observations that are not fitted [43, 44]. Negative values denote exothermic absorption; initial liquid, final liquid and pure-CO_2 feed use the same temperature and pressure.

The Kim–Svendsen comparison deteriorates at 393.15 K (Table 9). At that temperature its finite-interval heat RMSE exceeds the 2026 model's on the common intervals; Table 9 gives both errors and the shared count. Both calculations include the inconsistent printed intervals in this common-row comparison; alternative source exclusions are reported separately in the supplement. Reaction-law extrapolation, approximated pressure histories and missing chemistry can affect the calculated heat, but this result does not identify a unique cause. The discrepancy prevents interpreting the common enthalpy formulation as demonstrated high-temperature caloric accuracy.

The heat-capacity comparison extends beyond the fitted Hilliard states, with the same feed-mass convention used for all mass-specific values. Its error and the source subsets are included in Table 9. The broader comparison supplies information on temperature dependence without establishing independent validation, since the data were inspected during development.

4.3.3. Density

Loaded-density observations constrain the same Helmholtz energy used for pressure and speciation (Fig. 6). Agreement with the fitted Amundsen and Hartono observations is closer than with the full source ranges, which include other compositions and conditions. Han densities remain outside the objective, while the excluded Amundsen observations remain visible in the source account rather than being removed from assessment. The Amundsen–Hartono difference at the excluded loading does not establish which source is more accurate.

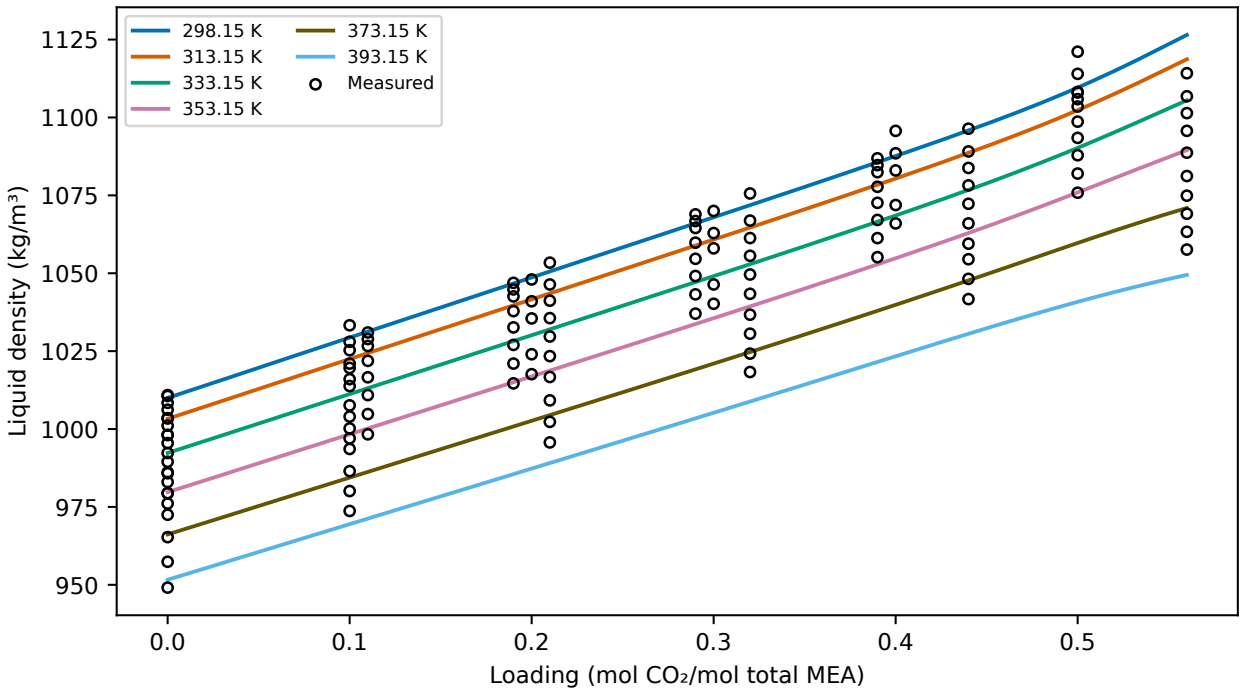


Figure 6: Liquid-density calculations without volume translation at nominal 30 wt% MEA. Lines use the labeled temperature at 0.1 MPa through 353.15 K and 0.7 MPa at higher temperatures; circles are the Amundsen, Hartono and Han observations at nominal 30 wt% [3, 45, 46]. Point comparisons use each source state; curves are not fits to every displayed observation. The full source comparison is reported in Table 9.

4.4. Other MEA concentrations

Calculations at 15 and 45 wt% MEA expose concentration dependence beyond the nominal-30-wt% pressure fit (Fig. 7). The pressure deviations in Table 9 remain substantial at both concentrations, and the parameter reduction does not remove this weakness. These observations are not fitted but informed the adopted choice, not independent tests of transfer. No additional concentration-specific parameters are estimated here.

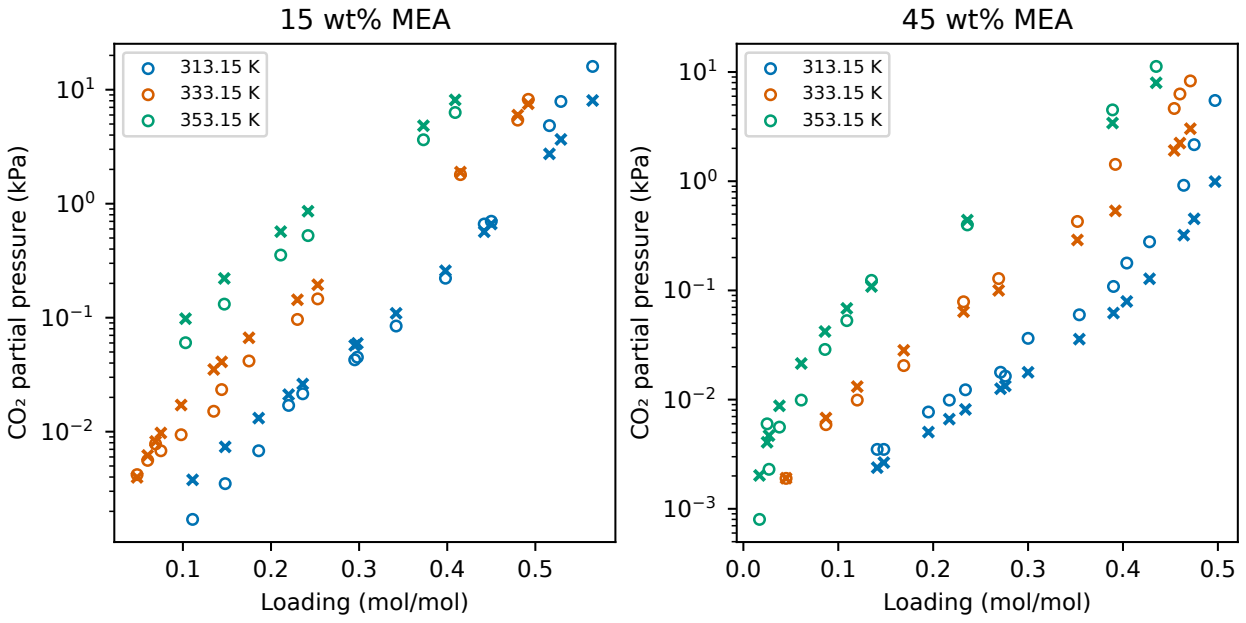


Figure 7: Measured pressures (open circles) and F2 calculations (crosses) at the same Aronu states for 15 and 45 wt% MEA [37]. Colors identify temperature. Each cross uses the source's composition, loading and temperature; no interpolated model curve is implied. These observations are not fitted and were seen during parameter selection.

4.5. Parameter choices and conditional uncertainty

Tables 7 and 8 identify the adopted species and interaction values; the reaction laws are in Table 1. Table 6 gives the measured cost of each cumulative restriction on the same observations and weights. F0 allows reaction-law shifts and separate amine-ion Born diameters; F1 fixes R2 and R5 to literature, and F2 additionally ties those Born diameters to segment diameters. F3 also fixes the carbamate–water interaction and $MEAH^+$–water temperature slope to zero. F4 fixes the water–bicarbonate and water–carbonate laws to sodium-salt values and frees $MEAH^+$–bicarbonate interaction; F4-0 fixes that new interaction to zero, whereas F5 instead adds zero $MEAH^+$–carbamate interaction to F4. The cumulative changes quantify the cost of the selected assumptions, not the isolated physical effect of each restriction.

Table 6
Cumulative restrictions and their measured cost on identical observations. P: 47 fitted pressures; U: 112 pressures not fitted; L: 47 fitted loaded densities. Deviations are AARD (percent). No across-role ranking is implied.

Set	Parameters	On bounds	C	P	U	L
F0	16	6	148.511	11.79	23.82	0.162
F1	13	2	150.532	11.96	24.18	0.1463
F2	11	1	150.763	11.64	24.89	0.1523
F3	9	1	169.138	13.98	33.16	0.2027
F4	9	2	174.451	15.46	38.48	0.2129
F4-0	8	1	176.049	15.59	38.53	0.2065
F5	8	3	186.615	20.09	39.7	0.2281

Only the $MEAH^+$–$MEACOO^-$ interaction is at a bound for F2; its lower-bound value doubles the unlike-pair dispersion energy under Eq. (3). The other parameters have conditional standard errors listed beside their values in Table S6. The water–bicarbonate interaction and the opposing amine-ion size changes are weakly determined; their large conditional errors preclude identifying the fitted values as unique molecular properties. Correlated covariance propagation describes local pressure sensitivity, while inward bound scans test short feasible changes with the other

Table 7
Segment, association and Born parameters of the adopted parameter set. Sources identify fitted quantities and transferred inputs; Table S1 gives eight-significant-figure values and the remaining inputs. A dash denotes an unused input.

Species	m^{seg}	σ (Å)	ϵ/k_B (K)	ϵ^{AB}/k_B (K)	κ^{AB}	d^{Born} (Å)	Source
CO_2	2.0729	2.7852	169.21	878.3249	0.0376	–	Segments: [21, Table 2]; association: water fit (energy/2)
MEA	3.0353	3.0435	277.174	2586.3	0.0363	–	Segments, energy: [5, Table 2]; association volume converted here
H_2O	1.2528	$\sigma_w(T)$	432.3788	1756.6498	0.0376	–	Water fit
$MEAH^+$	1	4.8558	208.7356	–	–	4.8558	Joint fit; segment count assumed
$MEACOO^-$	1	5.3480	179.8579	–	–	5.3480	Joint fit; segment count assumed
HCO_3^-	1	2.9296	70	–	–	3	Size, energy: [11, Table 4]; Born: assumed
CO_3^{2-}	1	2.4422	249.26	–	–	3	Size, energy: [11, Table 4]; Born: assumed
H_3O^+	1	3.4654	500	–	–	1.218	Size, energy transferred: [11, Table 4]; Born: assumed
OH^-	1	2.0177	650	–	–	3.0811	Size, energy: [11, Table 4]; Born: assumed

Table 8
Nonzero or fitted binary interactions, $k_{ij}(T) = k_{ij}(313.15\,\mathrm{K}) + b_{ij}(1/T - 1/313.15\,\mathrm{K})$. All unlisted unlike-charge or neutral pairs have $k_{ij} = 0$; same-sign ion dispersion is excluded. An absent slope is zero. Sources of fixed inputs are in Section S1.

Pair	$k_{ij}(313.15\,\mathrm{K})$	b_{ij} (K)	Source
CO_2–H_2O	−0.0464	−35.6651	CO_2–water fit
MEA–H_2O	−0.0889	−6.3627	Joint fit
H_2O–$MEAH^+$	−0.3577	96.1747	Joint fit
H_2O–$MEACOO^-$	−0.0661	–	Joint fit
H_2O–HCO_3^-	0.7650	–	Joint fit
$MEAH^+$–$MEACOO^-$	−1	–	Joint fit
H_2O–CO_3^{2-}	−0.25	–	Assumed
H_2O–H_3O^+	0.25	–	Assumed
H_2O–OH^-	−0.25	–	Assumed

parameters refitted. Neither calculation includes model discrepancy, reaction-law uncertainty, campaign dependence or uncertainty of a phase boundary.

4.6. Comparison with published models

Table 9 compares the adopted formulation with the 2026 Polley–Hedengren parameters on identical observations [14]. The predecessor's equation-of-state calorics are calculated here from its published thermodynamic specification; its empirical column enthalpy correlations are not substituted. The comparison shows gains in pressure and density agreement alongside remaining and worsened quantities, including high-temperature heat and the more concentrated pressure series. Because the fitting roles differ, these values do not define an across-role model ranking. The predecessor's published fitting account does not establish exact membership for every pressure and species row; its mixed-role labels preserve that uncertainty rather than asserting all rows were fitted.

Table 9

Identical-row comparisons with Polley–Hedengren (2026). Columns disclose each model's data role; they do not provide an across-role model ranking. For the 2026 species ratios and carbonate shares, the measurements informed an earlier ion fit despite not entering the reported formulation objective. Heat RMSE is kJ/mol CO_2; density RMSE is kg/m^3.

Quantity; metric	n	2026 role	F2 role	2026	F2
Six-source pressures; AARD (%)	159	fitted / not fitted	47 fitted / 112 not fitted	52.341	20.972
47 F2 fitted pressures; AARD (%)	47	fitted / not fitted	fitted	68.892	11.642
112 F2 not fitted pressures; AARD (%)	112	fitted / not fitted	not fitted	45.396	24.887
Species fractions; RMSE (mole fraction)	131	fitted / not fitted	101 fitted / 30 not fitted	0.0076606	0.0030596
Species ratios; AARD (%)	6	not fitted	fitted	155.37	17.067
Jakobsen carbonate share; AARD (%)	10	not fitted	not fitted; previously assessed	540.48	95.011
Vinjarapu et al. (2024) heats; RMSE (kJ/mol CO_2)	8	not fitted	fitted	2.2837	1.979
Arcis et al. (2011) heats; RMSE (kJ/mol CO_2)	14	not fitted	not fitted	9.1282	9.0263
Amundsen et al. (2009) loaded; RMSE (kg m^{-3})	68	not fitted	fitted / not fitted	142.95	6.666
Amundsen et al. (2009) CO_2-free; RMSE (kg m^{-3})	35	not fitted	fitted / not fitted	6.2347	2.4407
Hartono et al. (2014) loaded; RMSE (kg m^{-3})	40	not fitted	fitted / not fitted	153.08	2.0465
Han et al. (2012) all; RMSE (kg m^{-3})	67	not fitted	not fitted	135.27	4.8966
Four Hilliard heat capacities; AARD (%)	4	not fitted	fitted	11.411	2.8348
Broad heat capacity; AARD (%)	51	not fitted	2 fitted / 49 not fitted	11.395	2.5397
Weiland1997 comparison; AARD (%)	6	not fitted	not fitted	9.2583	4.8241
Concepcion2023 comparison; AARD (%)	3	not fitted	not fitted	14.386	2.1739
Wagner pressures; AARD (%)	23	not fitted	not fitted	33.44	25.033
Aronu 15 wt% pressures; AARD (%)	33	comparison only	not fitted	90.262	45.08
Aronu 45 wt% pressures; AARD (%)	37	comparison only	not fitted	32.128	45.859
Kim heat 313.15 K; RMSE (kJ/mol CO_2)	27	fitted / not fitted	not fitted	5.8021	4.9771
Kim heat 353.15 K; RMSE (kJ/mol CO_2)	26	fitted / not fitted	not fitted	10.178	9.612
Kim heat 393.15 K; RMSE (kJ/mol CO_2)	20	not fitted	not fitted	29.489	31.774

The Ma'mun observations provide an exact published-data match (Table 10) [40]. On the identical rows, pressure AARD worsens from F0 to F2, as the table shows. Zhang fitted these data, whereas Baygi and Najafloo describe ternary calculations outside their fitting objectives [1, Table 9][5, 6, Table 5]. Najafloo's detailed methods and conclusion differ in their fitting description, and equivalence of its pressure AAD formula is inferred rather than explicitly defined. Published aggregate errors on other datasets are not converted into common-row comparisons when their row selection or property basis cannot be reconstructed.

Table 10
Ma'mun pressure observations on identical rows. Published deviations and parameter counts refer to the source estimation stage. Najafloo's AAD definition is inferred equivalent; its conclusion and detailed fitting description differ. F0 is a comparison parameter set.

Model	n	Role	Parameters	Deviation (percent)
Baygi and Pahlavanzadeh (2015) PC-SAFT	19	not fitted	6	21.03
Najafloo and Zarei (2018) SAFT-HR	19	not fitted	5	39.96
Zhang et al. (2011) eNRTL	19	fitted	12	13.5
F0	19	not fitted	16	19.43
F2	19	not fitted	11	33.47

4.7. Scope and limitations

The adopted set is a calibration with calculations at fixed parameters against data seen during development; no comparison establishes independent validation. Its uncertainty is conditional on the fit, and its phase behavior is supported by finite searches rather than a global proof. Concentration-dependent pressures, carbonate shares and the highest-temperature heat remain inaccurate despite improved agreement with other properties. R1–R3 have a source-correlation range rather than verified measurement coverage, and R4/R5 and the unloaded dielectric law are extrapolated beyond their direct source constraints. Missing pressure histories, source inconsistencies, observable approximations, omitted carrier gas and omitted oxazolidone further limit interpretation. No absorber or stripper operating prediction is qualified by the property comparisons alone.

5. Conclusions

An ePC-SAFT calibration with literature reaction laws and tied amine-ion Born diameters describes pressure, speciation, heat, heat capacity and density through one Helmholtz energy without detected additional-liquid instability in the specified finite searches. The reduced parameter freedom has a measured cost, including worse agreement with the Ma'mun pressures, while concentration transfer, carbonate partition and high-temperature absorption heat remain inaccurate. The false split of the unpenalized comparison fit reflects opposing energy terms with a prominent dispersion change; the evidence does not identify one microscopic cause.

These are calibration and fixed-parameter comparisons against data seen during development, not independent validation. Local uncertainty remains conditional on the fitted model and inputs, and finite searches do not establish global stability. The common thermodynamic property calculation is useful for consistent assessment, but its remaining discrepancies preclude an unqualified process-use claim.

Table 11 specifies exactly the shared coefficient set held fixed in Part II; the associated fixed fields, topology and standard states are defined in the model section and supplement. It includes water, CO_2, hydronium, hydroxide, bicarbonate and carbonate, their shared interactions and R1–R3 laws. The poorly determined water–bicarbonate interaction, assumed dielectric behavior in loaded solutions and source-correlation extrapolation are the shared inputs most likely to limit transfer. Part II assesses the consequences with other amines rather than assuming that MEA agreement establishes their accuracy.

Table 11
Exact shared coefficient set held fixed in Part II, grouped by physical quantity. Water, CO_2, H_3O^+, OH^-, HCO_3^-, CO_3^{2-}, their shared interactions and R1–R3 laws are included. Topology, fixed fields and standard states also transfer as specified in the text.

Parameter	Value	Unit
CO_2 dispersion energy/k	169.21	kelvin
CO_2 segment count	2.0729	1
CO_2 segment diameter	2.7852	Å
CO_2 solvation factor	1	1
CO_2 dielectric volume	1×10^{-8}	liter / mole
CO_2 dielectric temperature	1403	kelvin
water dispersion energy/k	432.3788131	kelvin
water segment count	1.252753838	1
water solvation factor	1.5	1
water dielectric volume	0.3777	liter / mole
water dielectric temperature	1403	kelvin
HCO_3^- Born diameter	3	Å
HCO_3^- Debye–Hückel diameter	2.9296	Å
HCO_3^- dispersion energy/k	70	kelvin
HCO_3^- packing diameter	2.578048	Å
HCO_3^- segment count	1	1
HCO_3^- segment diameter	2.9296	Å
HCO_3^- solvation factor	1	1
CO_3^{2-} Born diameter	3	Å
CO_3^{2-} Debye–Hückel diameter	2.4422	Å
CO_3^{2-} dispersion energy/k	249.26	kelvin
CO_3^{2-} packing diameter	2.149136	Å
CO_3^{2-} segment count	1	1
CO_3^{2-} segment diameter	2.4422	Å
CO_3^{2-} solvation factor	1	1
H_3O^+ Born diameter	1.218	Å
H_3O^+ Debye–Hückel diameter	3.4654	Å
H_3O^+ dispersion energy/k	500	kelvin
H_3O^+ packing diameter	3.049552	Å
H_3O^+ segment count	1	1
H_3O^+ segment diameter	3.4654	Å
H_3O^+ solvation factor	1	1
OH^- Born diameter	3.081076894	Å
OH^- Debye–Hückel diameter	2.0177	Å
OH^- dispersion energy/k	650	kelvin
OH^- packing diameter	1.775576	Å
OH^- segment count	1	1
OH^- segment diameter	2.0177	Å
OH^- solvation factor	1	1
HCO_3^-–H_3O^+ k_{ij}	0	1
HCO_3^-–CO_2 k_{ij}	0	1
CO_2–CO_3^{2-} k_{ij}	0	1
CO_2–H_3O^+ k_{ij}	0	1
CO_2–OH^- k_{ij}	0	1
CO_2–water k_{ij}	−0.04641690112	1
CO_2–water k_{ij} temperature slope	−35.66505221	kelvin
CO_3^{2-}–H_3O^+ k_{ij}	0	1
H_3O^+–OH^- k_{ij}	0	1
HCO_3^-–water k_{ij}	0.7650000234	1
CO_3^{2-}–water k_{ij}	−0.25	1

Parameter	Value	Unit
H_3O^+–water k_{ij}	0.25	1
OH^-–water k_{ij}	−0.25	1
R1–R3 laws	Table 1	1

Data and Code Availability

The electronic data comprise processed observations, parameter values, numerical results and analysis and presentation code under the MIT license; third-party measurements retain their source terms. Deposit identifier: *[Zenodo DOI supplied in the final submission]*. A software index in the deposit specifies versions, checksums, omitted dependencies and calculation commands. The adopted MEA results use the author's ePC-SAFT implementation (Python package `epcsaft`, source commit `391dfe5d`); the comparison parameter sets are evaluated with that same implementation, as specified in the electronic data. Referee access to the compiled ePC-SAFT library: *[access arrangement supplied in the final submission]*.

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CRediT Author Statement

Tanner W. Polley: Conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing–original draft, writing–review and editing, and visualization.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this manuscript, OpenAI Codex and Claude (Anthropic) were used to draft and revise manuscript text, check computational consistency, and assist with bibliography checking and correction. The author directed the study, selected the model and parameters, and takes responsibility for the scientific judgments and final text.

Nomenclature

Symbol	Meaning	Unit	First equation
A, a^{res}	Extensive and reduced residual Helmholtz energy	J; 1	1
n, n_i	Total and species amount	mol	1
R, k_B	Gas and Boltzmann constants	J/(mol K); J/K	1
T, T_{ref}	Temperature and reference temperature	K	2
$m_i^{\mathrm{seg}}, \bar{m}$	Segment count and mixture average	1	6
x_i, y_i	True-species liquid and vapor mole fraction	1	6; text Sect. 2

Symbol	Meaning	Unit	First equation
$\sigma_i, \sigma_{ij}, d_i$	Segment, pair and packing diameter	Å	2
$\epsilon_i, \epsilon_{ij}$	Segment and pair dispersion energy	J	3
k_{ij}, b_{ij}	Interaction intercept and temperature slope	1; K	3, 4
$\epsilon^{AB}, \kappa^{AB}$	Association energy and volume factor	J; 1	5
ρ, ρ_N	Molar and molecular number density	mol/m^3; m^{-3}	8
ζ_n, η	Hard-sphere moments; packing fraction	m^{n-3}; 1	8
g_{ij}^{hs}	Hard-sphere contact value	1	8
I_1, I_2, C_1	Dispersion integrals and compressibility factor	1	9, 10
a_k, b_k	Dispersion polynomial coefficients	1	11
X_A, Δ^{AB}	Unbonded site fraction and bond strength	1; m^3	12, 13
e, ε_0	Elementary charge and vacuum permittivity	C; F/m	14
$z_i, \mathcal{I}$	Charge number and ion set	1	14
κ_D, χ_i	Inverse screening length and finite-size factor	m^{-1}; 1	14, 15
$\varepsilon_{\mathrm{bulk}}, \varepsilon_{\mathrm{ion}}$	Bulk and ion-region relative permittivity	1	14, 16
d_i^{Born}, D_i	Born and outer-shell diameter	Å	16
$c_{\mathrm{shell}}, c_{\mathrm{dielectric}}$	Shell and saturation coefficients	1	16, 17
f_i, f_{mix}	Solvation factor and mixture average	1	17
$d_{V,i}, d_{T,i}$	Dielectric-volume and temperature coefficients	L/mol; K	18
$n_s, V, \mathcal{S}$	Neutral-solvent amount, phase volume and neutral set	mol; m^3	18
ν_{ri}, K_r	Reaction stoichiometry and equilibrium constant	1	19
a_i, ϕ_i, μ_i	Activity, fugacity coefficient and chemical potential	1; 1; J/mol	20; text Sect. 2
$P, P^\circ, p_{\mathrm{CO_2}}$	Total, reference and partial pressure	Pa	20; text Sect. 2
Z, M_w, m°	Compressibility, water molar mass, standard molality	1; kg/mol; mol/kg	20
$K_{m,r}, K_{x,r}$	Molality- and mole-fraction-basis constant	1	21
$A_r, B_r, C_r, D_r, \Delta_r$	Reaction coefficients and standard-state offset	1; K; 1; K^{-1}; 1	22
H, C_P, c_P	Enthalpy, extensive and mass heat capacity	J; J/K; J/(kg K)	23; text Sect. 2
α, w_{MEA}	Loading and unloaded-solvent mass fraction	mol/mol; kg/kg	text Sect. 3.1
$\Delta h, h^{\mathrm{feed}}$	Absorption heat and feed molar enthalpy	J/mol	24
$u_\rho, u_\alpha, g_j, s_\rho$	Density/loading scale inputs and density secant	kg/m^3; mol/mol; kg/m^3; kg/m^3	25
$\boldsymbol{\theta}, \boldsymbol{C}, r_j, w_j, s_j$	Parameters, objective, residual, weight and scale	various; 1; 1; 1; observable	26
$\boldsymbol{J}_I, s_C^2, I, N$	Interior weighted Jacobian, variance factor, interior set, count	various; 1; 1; 1	28
$g, D, \boldsymbol{w}$	Molar Gibbs energy, reduced distance and trial fraction	J/mol; 1; 1	29
$\boldsymbol{B}, \boldsymbol{\lambda}$	Conservation matrix and reduced multipliers	1	29
$\tau_k, s_{\mathrm{stab}}, w_{\mathrm{stab}}, \boldsymbol{C}_{\mathrm{pen}}$	Descending distance, penalty scale/weight and cost	1	30

Subscripts i, j identify species, r reactions, w water and k polynomial or search indices; superscripts res, ig, hs, hc, disp, assoc, DH and Born identify residual, ideal-gas, hard-sphere, hard-chain, dispersion, association and electrostatic terms. Superscripts calc and obs mean calculated and observed; pure and ∞, w mean pure solvent and infinite dilution in water; $\circ$ denotes the source standard state. AARD: average absolute relative deviation; RMSE: root-mean-square error; MEA: monoethanolamine; MDEA: N-methyldiethanolamine; NMR: nuclear magnetic resonance; VLE: vapor–liquid equilibrium; eNRTL: electrolyte non-random two-liquid; PC-SAFT: perturbed-chain statistical associating fluid theory; SSM+DS: solvation-shell model with dielectric saturation.

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